

Downlink and Data Management (DDM) CFDP Dashboard Demo for Europa Clipper Operations

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About Me

- B.S. in Aerospace Engineering from MIT (2023)
- My time at the Jet Propulsion Laboratory, California Institute of Technology
 - Summer internship + APX semester on Psyche (2021)
 - Summer internship on MSR SRL (2022)
 - Started full time on Europa Clipper in August 2023
 - Brief work on GRACE-C and GDS MITD while still supporting Clipper
- Hobbies:
 - Roadtrips
 - Hiking
 - Video Games

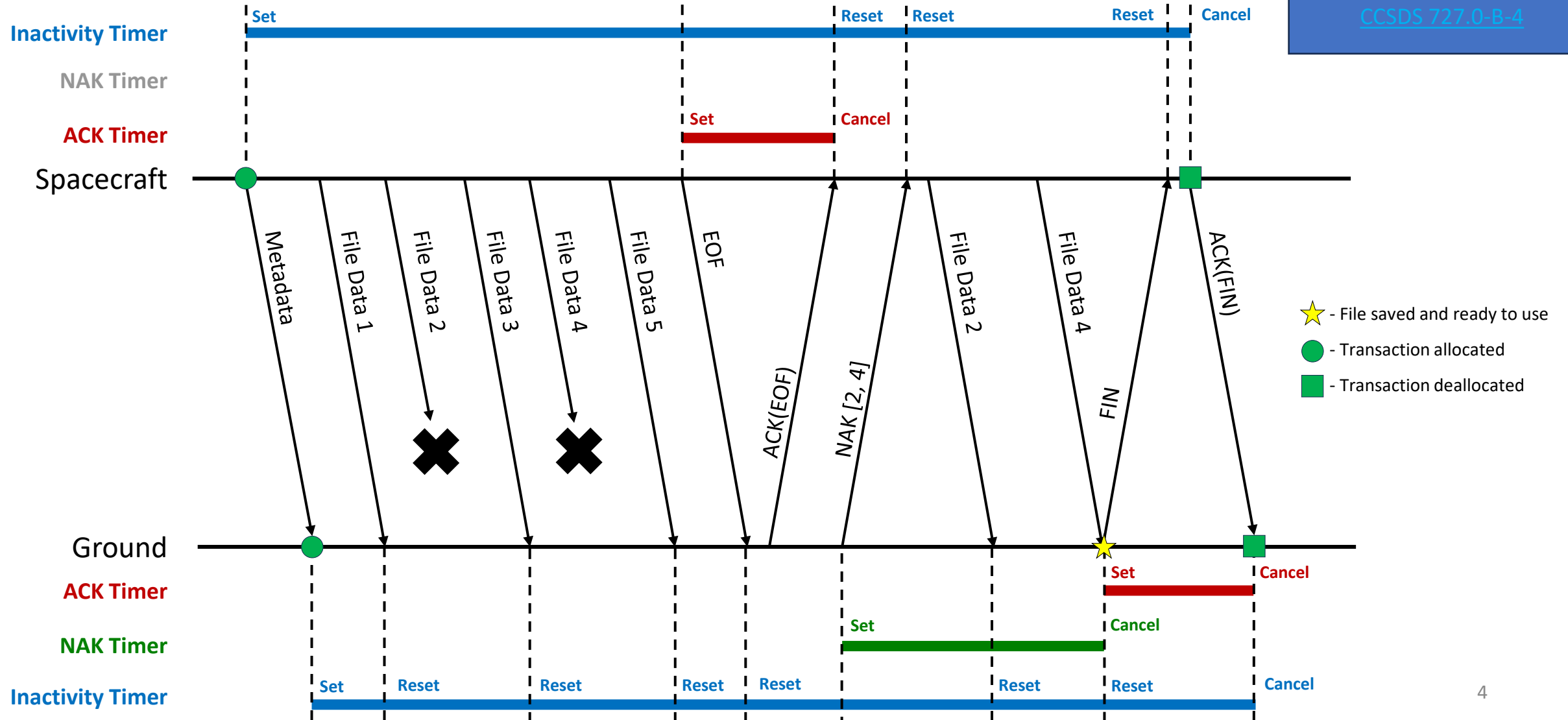


Class 2 CFDP and How It Helps Europa Clipper

- The Class 2 CCSDS File Delivery Protocol (CFDP) is a reliable transport protocol that is handled at each end point of the communication link
 - In Clipper's case, the end points are the Ground Data System (GDS) and the Flight System (FS)
- Reliable CFDP utilizes Control Maintenance Messages to indicate:
 - End of File – Initial transmit is complete
 - Negative Acknowledgement – File segments are missing and are requested for retransmit
 - Finished – The receiving entity has successfully received all metadata and file data
- Europa Clipper utilizes Class 2 CFDP to more efficiently use the available downlink bandwidth, trigger automated retransmits, and manage the on-board file system
 - Data product downlink can be enabled during periods of track coverage that may be noisier, allowing more overall bandwidth utilization as opposed to holding DP downlink until we have confidence the telecom noise has decreased.
 - The no-error Finished indications allow the data product sending entity (The Flight System) to safely delete data products automatically, keeping the on-board file system utilization low

CFDP Downlink Diagram (with Drops)

Custom visualization of the public CCSDS File Delivery Protocol specification in [CCSDS 727.0-B-4](#)



How do we monitor CFDP in Operations?

- Spacecraft Telemetry
 - Continuous / Channelized – Engineering health and accountability data (EHA)
 - Discrete – Event Records (EVR)
- DDM CFDP Dashboard (**today's focus**)
- Requesting ground-specific context/configuration from the Ground Data Systems Operations (GDSO) team

DDM CFDP Dashboard – Problem Statement

- Problem: As the Operations team prepared for Launch via ORTs, DDM recognized that our visibility into CFDP would be restricted compared to V&V
 - The easy-to-view dashboard was restricted to only GDS because it also manages the ground configuration
- I was requested to develop a way to visualize the live progress of active file downlinks beyond the AMPCS product status monitor

AMPCS Product Status Monitor

Product Status ⓘ			
Status	Type	# Recvd	DVT SCLK
	DP_REASON_BIN_4	630	0495201502.4179008
	DP_REASON_BIN_4	630	0495201505.0117188
	DP_REASON_BIN_4	630	0495201501.0898438
	DP_REASON_BIN_4	630	0495201505.6210938
	DP_REASON_BIN_4	630	0495201503.0468750
	DP_REASON_BIN_4	630	0495201501.7851563
	DP_REASON_BIN_4	630	0495201509.5429688
	DP_REASON_BIN_4	630	0495201514.0976563
	DP_REASON_BIN_4	630	0495201512.1367188
	DP_REASON_BIN_4	630	0495201511.5039063
	DP_REASON_BIN_4	630	0495201510.8710938
	DP_REASON_BIN_4	630	0495201508.9101563
	DP_REASON_BIN_4	630	0495201506.9492188
	DP_REASON_BIN_4	630	0495201506.3437500
	DP_REASON_BIN_4	630	0495201508.3046875
	DP_REASON_BIN_4	630	0495201497.8632813
	DP_REASON_BIN_4	630	0495201499.2148438
In Progress	DP_REASON_BIN_4	310	0495201516.6640625
Complete	DP_REASON_BIN_4	630	0495201510.1757813
	DP_REASON_BIN_4	630	0495201504.3789063
	DP_REASON_BIN_4	630	0495201498.4960938
	DP_REASON_BIN_4	630	0495201497.2539063
	DP_REASON_BIN_4	630	0495201499.8242188
	DP_REASON_BIN_4	630	0495201507.5820313

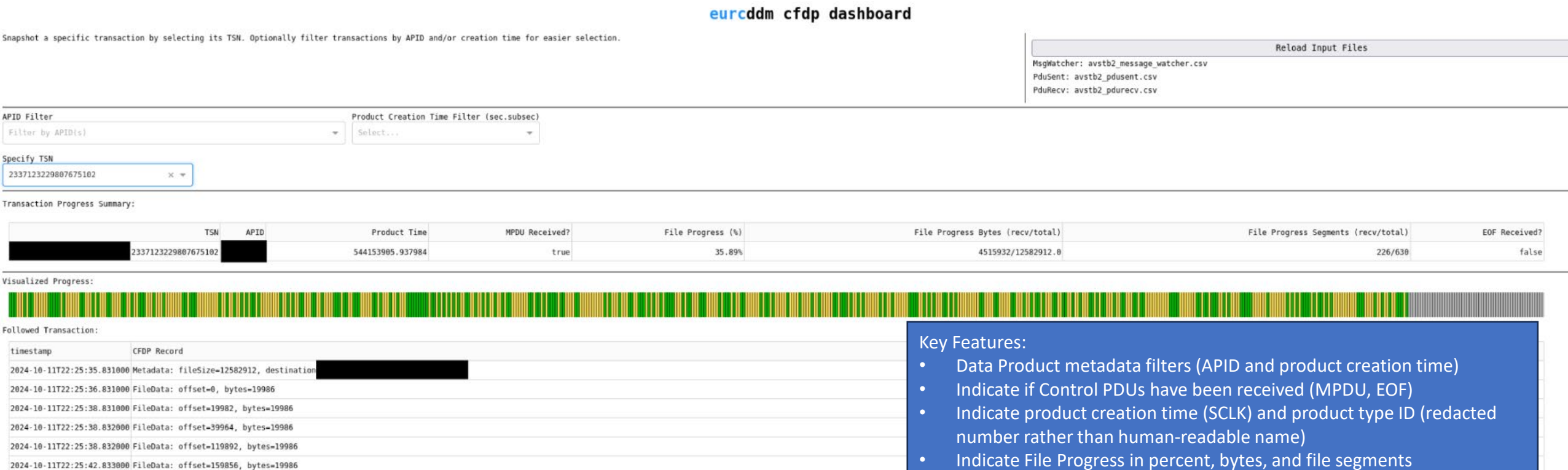
This is the standard multi-mission product status monitor which

- Indicates In Progress / Complete / Partial status,
- Data product type and spacecraft clock (SCLK) time,
- Number of parts received, but no indication of the total progress (% complete or volume)

This view is most helpful for high-level assessment across DPs (e.g. which DPs are complete)

DDM CFDP Dashboard – Launch Readiness

Immediate solution: A dashboard with barebone filters, a transaction summary table, static status bar, and ordered PDU printout



Key Features:

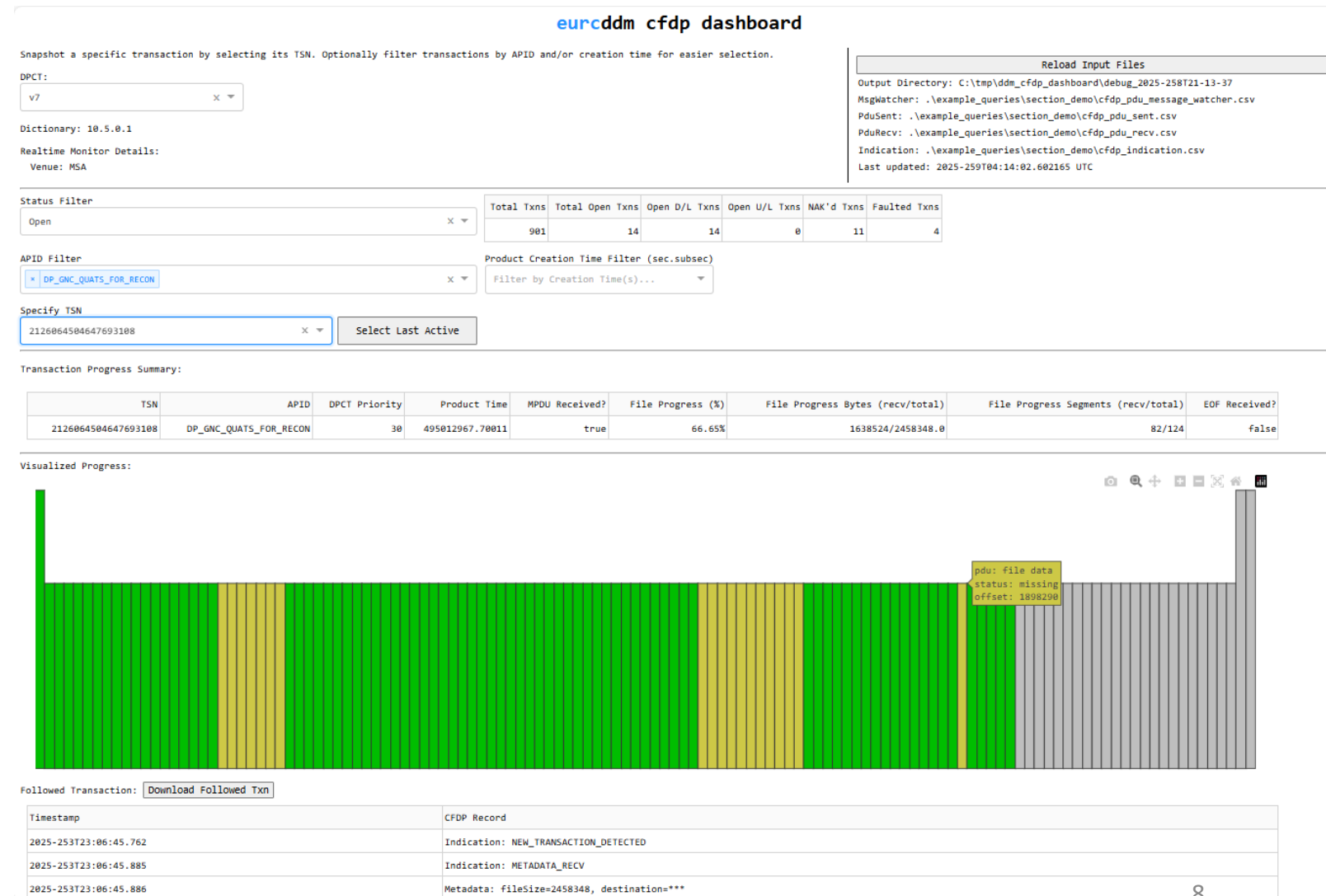
- Data Product metadata filters (APID and product creation time)
- Indicate if Control PDUs have been received (MPDU, EOF)
- Indicate product creation time (SCLK) and product type ID (redacted number rather than human-readable name)
- Indicate File Progress in percent, bytes, and file segments
- Visually indicate if file data PDUs are missing (yellow), received (green), or waiting (grey) in a **static** plot
- Displayed transaction PDUs by timestamp

Pre-Launch DDM CFDP Dashboard Minimum Viable Product (sandbox example)

DDM CFDP Dashboard – Today

I've continued to iterate on the dashboard based on user feedback and general QoL, including:

- Select Last Active button
- Interactive plot with MPDU, ACK(Eof) and FIN receive status visually indicated
- Overall status summaries and status filtering
- Human-readable APID
- Ingesting downlink priority configuration table (DPCT) to report downlink priority
- Additional record types in the time-based transaction printout (Indications)
- Download button for the transaction printout



How the DDM CFDP Dashboard Works

Data Interfaces:

- The MariaDB that stores CFDP records via `chill_get_cfdp` queries to capture historical PDU sent, PDU received, and indication records
- The Java Messaging Service (JMS) topic bus via `chill_message_watcher` to inform live updates

Codebase:

- Python primarily leveraging Plotly Dash for the dashboard itself and Pandas for data processing
- Additionally utilizes some utility scripts that were produced during V&V and certified for use in Operations to:
 - Show human-readable names for data products
 - Abstract the `chill_get_cfdp` queries
 - Produce rich logging

Deployment:

- Maintained in a GitHub repository
- Cloned in the MSA and on the ITS machines, launched by users following the README.md
- I'm hoping to get the dashboard hosted on a dedicated server or machine to make it more accessible

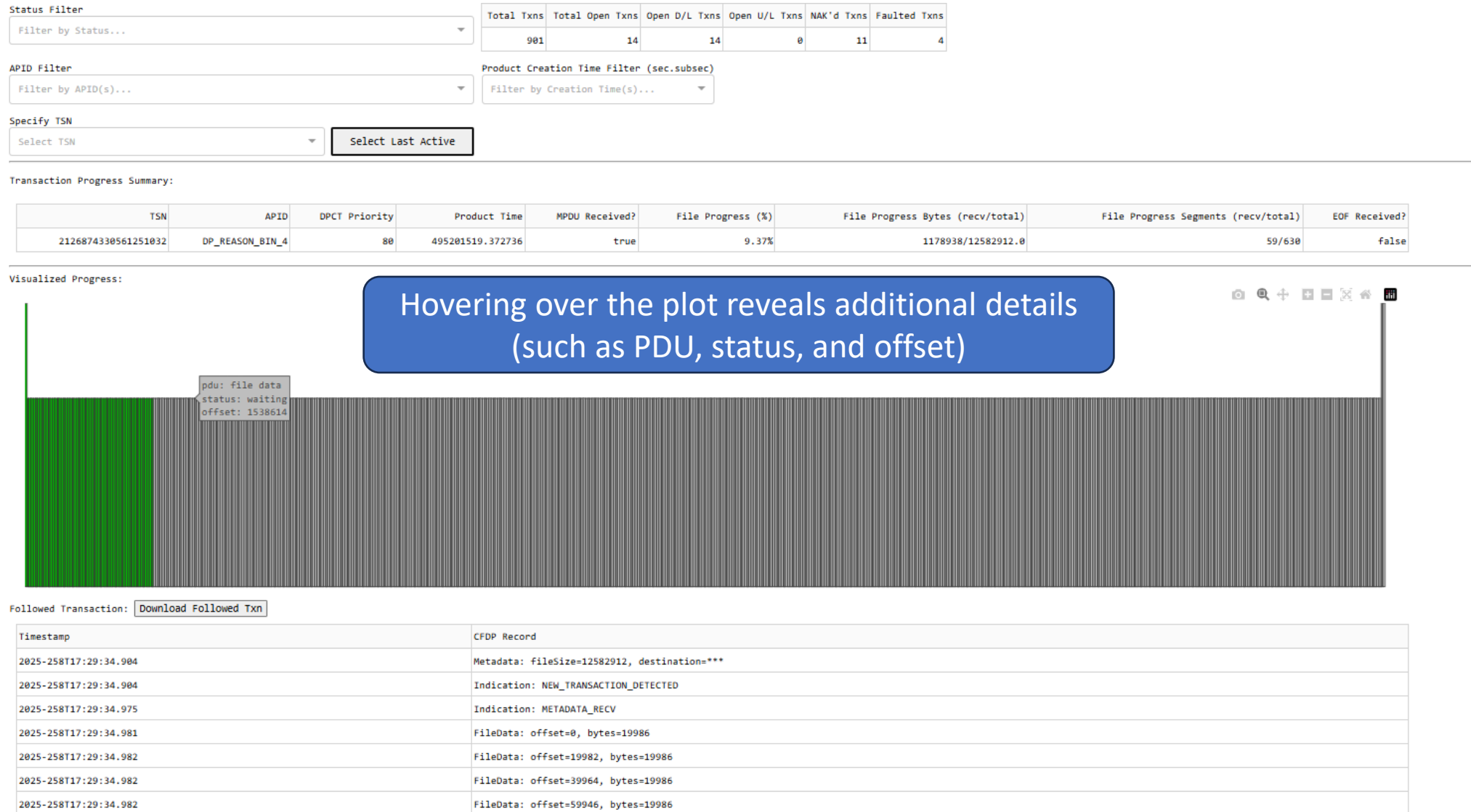
Examples

Disclaimer: The following examples are sandbox examples to highlight the dashboard capabilities by creating types of CFDP transactions

All examples use a modified version of the dashboard for use in this slide deck

DP Downlink in Progress (no drops)

REASON BIN 4 Data Product with no drops, and only about 10% downlinked



DP Downlink in Progress (with drops)

Filtered for Open transactions and DP_GNC_QUATS_FOR_RECON data products, leaving only a single TSN left in the “Specify TSN” options

Status Filter

Open

APID Filter

DP_GNC_QUATS_FOR_RECON

Specify TSN

2126064504647693108

Select Last Active

Total Txns

901

Total Open Txns

14

Open D/L Txns

14

Open U/L Txns

0

NAK'd Txns

11

Faulted Txns

4

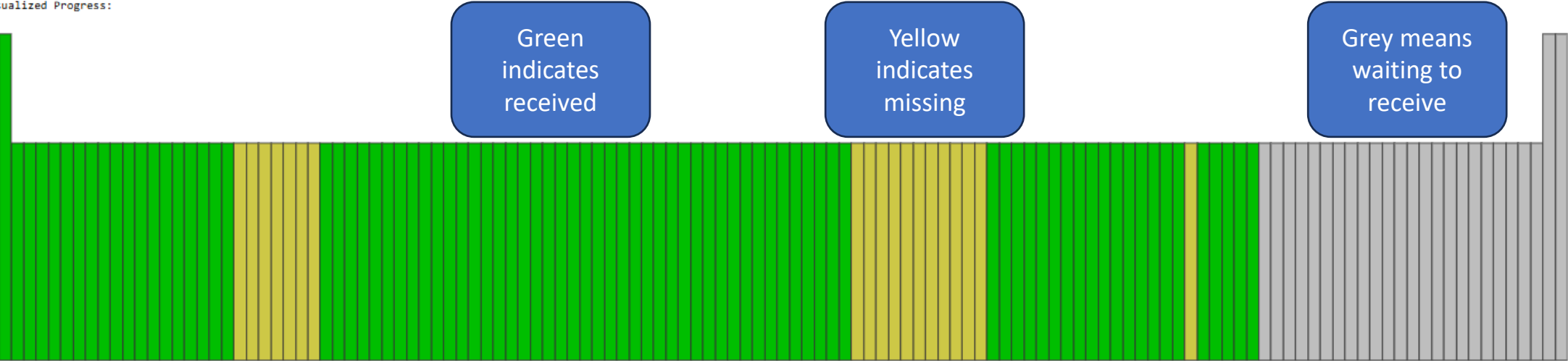
Product Creation Time Filter (sec.subsec)

Filter by Creation Time(s)...

Yellow is not bad! Missing data is nominal and handled seamlessly by Class 2 CFDP

TSN	APID	DPCT Priority	Product Time	MPDU Received?	File Progress (%)	File Progress Bytes (recv/total)	File Progress Segments (recv/total)	EOF Received?
2126064504647693108	DP_GNC_QUATS_FOR_RECON	30	495012967.70011	true	66.65%	1638524/2458348.0	82/124	false

Visualized Progress:

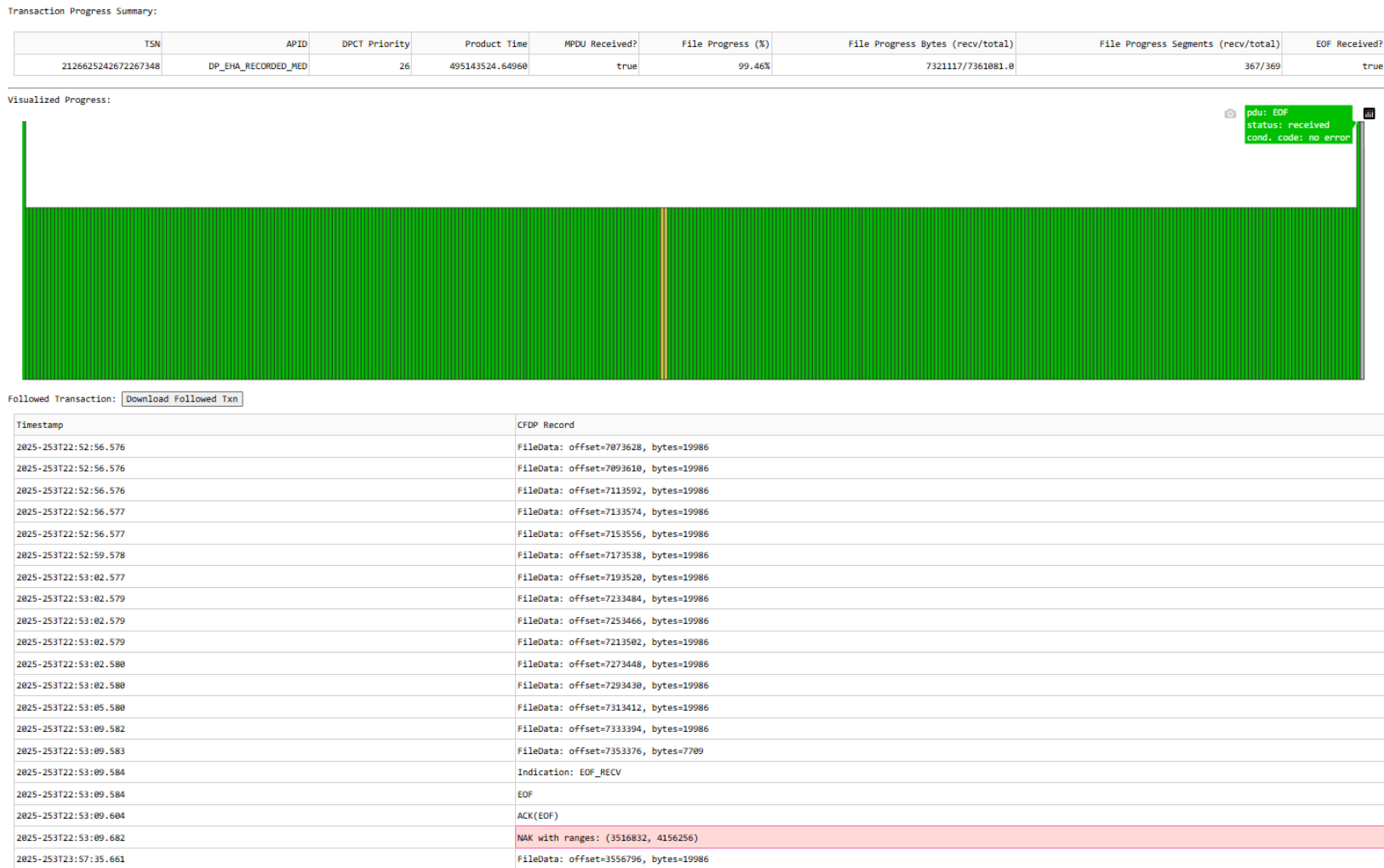


Followed Transaction: Download Followed Txn

Timestamp	CFDP Record
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DP with Issued Retransmit Request

This product had data dropped, requested retransmit via NAK after the EOF was received, and then started receiving retransmitted data. The snapshot was taken when only two PDUs were left to downlink from the initial NAK



Cancelled Transaction

This example transaction reached the ground Inactivity Detected fault before it completed, resulting in a cancelled transaction. If this were Flight, the data product would need to be manually retransmitted

Status Filter

Fault Condition

Total Txns

901

Total Open Txns

14

Open D/L Txns

14

Open U/L Txns

0

NAK'd Txns

11

Faulted Txns

4

APID Filter

Filter by APID(s)...

Product Creation Time Filter (sec.subsec)

Filter by Creation Time(s)...

Specify TSN

2126140260716126632

Select Last Active

Transaction Progress Summary:

TSN	APID	DPCT Priority	Product Time	MPDU Received?	File Progress (%)	File Progress Bytes (recv/total)	File Progress Segments (recv/total)	EOF Received?
2126140260716126632	DP_EHA_STATS	10	495030605.420418	true	62.6%	99910/159589.0	5/8	false

Visualized Progress:

pdu: ACK(FIN)

status: fault

cond. code: inactivity detected

Followed Transaction:

Download Followed Txn

Timestamp	CFDP Record
2025-253721:56:23.913	Indication: NEW_TRANSACTION_DETECTED
2025-253721:56:23.990	Indication: METADATA_RECV
2025-253721:56:23.991	Metadata: fileSize=159589, destination=***
2025-253721:56:23.999	FileData: offset=0, bytes=19986
2025-253721:56:24.000	FileData: offset=19982, bytes=19986
2025-253721:56:27.002	FileData: offset=39964, bytes=19986
2025-253721:56:30.005	FileData: offset=59946, bytes=19986
2025-253721:56:30.005	FileData: offset=79928, bytes=19986
2025-253721:56:30.134	FIN(Inactivity detected)
2025-253722:55:27.021	ACK(FIN)(Inactivity detected)
2025-253722:55:27.021	Indication: FAULT (INACTIVITY_DETECTED)

Finished Uplink Transaction

There are only two PDUs that the ground waits for during uplink CFDP, so the uplink progress is simpler, only showing the receive status of the ACK(EOF) and FIN PDUs

Status Filter

Filter by Status...

APID Filter

UPL_CFDP

Specify TSN

3314

Select Last Active

Product Creation Time Filter (sec.subsec)

Filter by Creation Time(s)...

Total Txns	Total Open Txns	Open D/L Txns	Open U/L Txns	NAK'd Txns	Faulted Txns
981	14	14	0	11	4

Transaction Progress Summary:

TSN	ACK(EOF) Received?	FIN Received?
3314	true	true

Visualized Progress:

ACK(EOF)

FIN

Followed Transaction:

Download Followed Txn

Timestamp	CFDP Record
2025-254T22:25:08.620	Indication: TRANSACTION
2025-254T22:25:08.621	Metadata: fileSize=437, destination=***
2025-254T22:25:08.634	FileData: offset=0, bytes=441
2025-254T22:25:08.642	EOF
2025-254T22:25:08.659	Indication: EOF_SENT
2025-254T23:20:32.209	ACK(EOF)
2025-254T23:20:35.211	FIN
2025-254T23:20:35.213	ACK(FIN)
2025-254T23:20:35.230	Indication: TRANSACTION_FINISHED

Failed Uplink Transaction

This uplink transaction triggered an internal CFDP fault (Filestore rejection) which is indicated in the PDU table or by hovering over the FIN. This resulted in a cancelled / failed uplink.

If this were in Flight, the Operations team would determine the root cause and reattempt uplink when it was safe to do so.

Status Filter

Fault Condition

APID Filter

UPL_CFDP

Specify TSN

3311

Select Last Active

Total Txns	Total Open Txns	Open D/L Txns	Open U/L Txns	NAK'd Txns	Faulted Txns
901	14	14	0	11	4

Transaction Progress Summary:

TSN	ACK(EOF) Received?	FIN Received?
3311	true	true

Visualized Progress:

ACK(EOF)

FIN

Followed Transaction:

Download Followed Txn

Timestamp	CFDP Record
2025-254T22:09:41.072	Indication: TRANSACTION
2025-254T22:09:41.073	Metadata: fileSize=1410, destination=***
2025-254T22:09:41.295	FileData: offset=0, bytes=1009
2025-254T22:09:41.446	FileData: offset=1005, bytes=409
2025-254T22:09:41.622	EOF
2025-254T22:09:41.631	Indication: EOF_SENT
2025-254T23:05:08.214	ACK(EOF)
2025-254T23:05:08.214	FIN(Filestore rejection)
2025-254T23:05:08.215	ACK(FIN)(Filestore rejection)
2025-254T23:05:08.233	Indication: TRANSACTION_FINISHED

Multi-Select Filters

Use case: Let's say I observed a set of data products created at a specific time and I'm interested in looking at the transactions just for specific APIDs (out of many)

- Setting the Product Creation Time filtered down options for APID Filter and for Specify TSN
- Selecting DPs in the APID Filter further filtered down options for Specify TSN to only 3 transactions

Status Filter

Filter by Status...

APID Filter

× DP_EVR_REC_WARNING_LOW

× DP_EVR_REC_CMD

× DP_EVR_REC_ACT_LOW

×

Specify TSN

Select TSN

2124621429996281924

2124621429996281928

2124621429996281930

Select Last Active

Total Txns	Total Open Txns	Open D/L Txns	Open U/L Txns	NAK'd Txns	Faulted Txns
901	14	14	0	11	4

Product Creation Time Filter (sec.subsec)

× 494676975.78086

Time	MPDU Received?	File Progress (%)	File Progress Bytes (recv/total)
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Appendix

Acronyms

- APX – Academic Part Time
- CCSDS – The Consultative Committee for Space Data Systems
- CFDP – CCSDS File Delivery Protocol
- DDM – Downlink and Data Management
- GDSO – Ground Data System Operations
- MITD – Multimission Integration, Test, and Deployment
- MSR SRL – Mars Sample Return, Sample Retrieval Lander
- ORT – Operational Readiness Test
- PDU – Protocol Data Unit
 - MPDU – Metadata PDU
 - FPDU – File Data PDU
 - EOF – End of File
 - ACK – Acknowledgement, typically followed by the type: ACK(EOF) / ACK(FIN)
 - NAK – Negative Acknowledgement
 - FIN – Finished
 - CMM – Control Maintenance Message (PDUs that are not MPDU or FPDU)

Resources

- CFDP Silver Book – [CCSDS 727.0-B-4](#)
- JPL-Internal Documents
 - Europa Clipper Flight Ground Interface Control Document (Rev D)
 - Europa Clipper Downlink and Data Management Functional Design Document (Rev L)
 - Europa Clipper Uplink and Command Functional Design Document (Rev A)